Performance benchmarks using the Rust Scientific Library (Russell)

Dr Dorival Pedroso

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We test the linear system $\boldsymbol{A}\,\boldsymbol{x} = \boldsymbol{b}$ where $\boldsymbol{A}$ is the coefficient matrix, $\boldsymbol{x}$ is the solution vector, and $\boldsymbol{b}$ is the right-hand side vector. The coefficient matrix is set with matrices from the SuiteSparse Matrix Collection. The right-hand side vector is filled with ones, i.e., we study the solution of

$$\boldsymbol{A}\,\boldsymbol{x} = \boldsymbol{1} \tag{1}$$

The relative error is calculated as

$$\texttt{RelativeError} = \frac{\max(|\boldsymbol{A}\,\boldsymbol{x} - \boldsymbol{1}|)}{\max(|\boldsymbol{A}| + 1)} \tag{2}$$

The tested real-valued matrices are:

1 `bwm2000` (Bai) – Brusselator wave model in transport interaction of chemical solutions (1992)
2 `rdb5000` (Bai) – Reaction-diffusion brusselator model (1994)
3 `Goodwin_040` (Goodwin) – Finite element, Navier-Stokes & other transport equations (2018)
4 `fp` (MKS) – 2-D Fokker Planck eqn, electron dyn. in external field (2006)
5 `xenon1` (Ronis) – Complex zeolite, sodalite crystals (2001)
6 `twotone` (ATandT) – Harmonic balance method (2001)
7 `Raj1` (Rajat) – Circuit Simulation Problem (2007)
8 `boyd2` (GHS_indef) – Optimization problem (2004)
9 `Goodwin_071` (Goodwin) – Finite element, Navier-Stokes & other transport equations (2018)
10 `darcy003` (GHS_indef) – Discretization using mixed FE of Darcy (2002)
11 `rma10` (Bova) – 3D CFD model, Charleston harbor (1997)
12 `helm2d03` (GHS_indef) – Helmholtz eq on a unit square (2004)
13 `stomach` (Norris) – 3D electro-physical model of a duodenum (2003)
14 `oilpan` (GHS_psdef) – Structural problem (2004)
15 `ASIC_680k` (Sandia) – Circuit simulation matrix (2006)
16 `tmt_unsym` (CEMW) – Electromagnetics problem (2008)
17 `Goodwin_127` (Goodwin) – Finite element, Navier-Stokes & other transport equations (2018)
18 `pre2` (ATandT) – Harmonic balance method (2001)
19 `marine1` (Martin) – Chemical oceanography; a marine nitrogen cycle inverse model (2018)
20 `torso1` (Norris) – Finite differences and boundary element, 2D model of torso (2003)
21 `atmosmodd` (Bourchtein) – CFD analysis of atmospheric models (2009)
22 `atmosmodl` (Bourchtein) – CFD analysis of atmospheric models (2009)
23 `memchip` (Freescale) – Circuit simulation problem (2010)
24 `Freescale1` (Freescale) – Circuit simulation problem (2008)
25 `rajat31` (Rajat) – Circuit simulation problem (2006)
26 `Transport` (Janna) – 3D finite element flow and transport (2012)
27 `inline_1` (GHS_psdef) – Structural problem, stiffness matrix (2004)
28 `PFlow_742` (Janna) – 3D pressure-temperature evolution in porous media (2014)
29 `Emilia_923` (Janna) – Geomechanical model for C02 sequestration (2011)
30 `dielFilterV2real` (Dziekonski) – FEM in electromagnetics (2011)
31 `Flan_1565` (Janna) – Structural problem, 3D model of a steel flange (2011)
32 `pres-cylin-3d-tet10-fine` (Pedroso) – FEM stiffness matrix of a pressurized cylinder (Tet10 with 1,711,464 DOF)
33 `pmsim-hex8-27` (Pedroso) – Global stiffness matrix of the stress-strain driver's 3x3x3 Hex8 block (27 elements, 157 unknowns after the essential boundary conditions)

The tested complex-valued matrices are:

1 `mhd1280b` (Bai) – Alfven spectra in magnetohydrodynamics (1994)
2 `mplate` (Cote) – Vibro-acoustic problem (1997)
3 `RFdevice` (Rost) – Semiconductor device simulation (2007)
4 `vfem` (CEMW) – Electromagnetics; vector finite element (2008)
5 `fem_filter` (Lee) – FEM band-pass microwave filter 500MHz (2008)
6 `Chevron4` (Chevron) – Temporal freq domain seismic modeling (2012)
7 `mono_500Hz` (FreeFieldTechnologies) – 3D vibro-acoustic problem, aircraft engine nacelle (2008)
8 `kim2` (Kim) – 2D 676-by-676 complex mesh (2002)
9 `fem_hifreq_circuit` (Lee) – FEM Maxwell equations for hi-freq circuit (2009)
10 `dielFilterV3clx` (Dziekonski) – High-order vector finite element method in EM (2011)

Additional notes:

- Each problem is solved five times. The reported error is the maximum among runs. The reported computer time is the average without outliers.
- Total time includes initialization (memory allocation + symbolic factorization), numeric factorization, and solve.
- Column NNZ (number of non-zeros) corresponds to Pattern Entries reported by the SuiteSparse Matrix Collection.
- Column Sym shows if symmetry information was provided to the solver. An asterisk means positive definite information was also given.
- oom (out-of-memory) indicates that the symbolic factorization was terminated due to insufficient memory.
- n/a indicates that the run was not attempted, or that the Direct solver exceeded its time or memory budget.
- Bold values highlight significant results, such as large errors or high computation times.
- The Direct solver is pure Rust; it does not compute determinants, error estimates or condition numbers.

Information about the system:

```
--- OS ---
NAME="Omarchy"
VERSION_ID="4.0.0.r7352.gaba49d2"
KERNEL=7.2.8-10-omarchy

--- GPU ---
03:00.0 VGA compatible controller: Advanced Micro Devices, Inc. [AMD/ATI] Navi 48 [Radeon RX 9070/9070 XT/907

--- CPU ---
Architecture       : x86_64
CPU(s)             : 32
On-line CPU(s) list: 0-31
Model name         : 13th Gen Intel(R) Core(TM) i9-13900KF
Thread(s) per core : 2
Core(s) per socket : 24
Socket(s)          : 1
CPU(s) scaling MHz : 43%
CPU max MHz        : 5800.0000
CPU min MHz        : 800.0000
BogoMIPS           : 5990.40
L1d cache          : 896 KiB (24 instances)
L1i cache          : 1.3 MiB (24 instances)
L2 cache           : 32 MiB (12 instances)
L3 cache           : 36 MiB (1 instance)
NUMA node0 CPU(s)  : 0-31
Vulnerability L1tf : Not affected

--- Memory ---
MemTotal:       32608328 kB
MemFree:         6813980 kB
MemAvailable:   19621072 kB
SwapTotal:      65224484 kB
```

Information about the libraries:

```
UMFPACK: local (/usr/local/lib/umfpack)
MUMPS: 5.9.0
SuiteSparse: latest (from GitHub)
```

The results are given in Tables 1 and 2. The iterative solvers are compared with UMFPACK in Table 3.

Table 1: Calculations on Arch. Direct, UMFPACK and MUMPS. Real-valued matrices.

				Direct		UMFPACK		MUMPS	
Matrix	**Nrow**	**NNZ**	**Sym**	**Total Time**	**Relative Error**	**Total Time**	**Relative Error**	**Total Time**	**Relative Error**
bwm2000	2,000	7,996	No	1.106ms	1.48×10^{-15}	750.259µs	5.30×10^{-16}	6.054ms	7.50×10^{-16}
rdb5000	5,000	29,600	No	39.034ms	1.01×10^{-14}	13.569ms	1.93×10^{-15}	9.946ms	9.40×10^{-13}
Goodwin_040	17,922	561,677	No	448.636ms	9.71×10^{-12}	107.091ms	6.62×10^{-13}	104.564ms	1.65×10^{-11}
fp	7,548	848,553	No	903.691ms	4.88×10^{-20}	154.476ms	2.57×10^{-21}	168.911ms	1.16×10^{-20}
xenon1	48,600	1,181,120	No	55.233s	1.19×10^{-38}	598.244ms	5.55×10^{-40}	365.443ms	5.86×10^{-40}
twotone	120,750	1,224,224	No	n/a	n/a	337.133ms	1.13×10^{-14}	983.724ms	1.99×10^{-13}
Raj1	263,743	1,302,464	No	n/a	n/a	oom	oom	947.420ms	3.63×10^{-13}
boyd2	466,316	1,500,397	Yes	n/a	n/a	**1m18.248s**	2.86×10^{-13}	844.848ms	8.98×10^{-11}
Goodwin_071	56,021	1,797,934	No	n/a	n/a	426.060ms	2.77×10^{-12}	290.877ms	4.87×10^{-11}
darcy003	389,874	2,101,242	Yes	n/a	n/a	1.385s	6.95×10^{-12}	11.601s	1.88×10^{-10}
rma10	46,835	2,374,001	No	n/a	n/a	410.052ms	1.57×10^{-17}	169.806ms	1.35×10^{-16}
helm2d03	392,257	2,741,935	Yes	n/a	n/a	1.683s	2.37×10^{-13}	1.573s	2.75×10^{-10}
stomach	213,360	3,021,648	No	n/a	n/a	1.781s	5.13×10^{-16}	1.720s	1.49×10^{-15}
oilpan	73,752	3,597,188	Yes*	n/a	n/a	494.931ms	1.26×10^{-15}	305.492ms	2.05×10^{-15}
ASIC_680k	682,862	3,871,773	No	n/a	n/a	1.318s	2.20×10^{-11}	**1m19.532s**	8.09×10^{-11}
tmt_unsym	917,825	4,584,801	No	n/a	n/a	3.747s	5.96×10^{-8}	3.289s	1.43×10^{-7}
Goodwin_127	178,437	5,778,545	No	n/a	n/a	1.173s	6.03×10^{-12}	884.040ms	8.89×10^{-10}
pre2	659,033	5,959,282	No	n/a	n/a	oom	oom	5.102s	2.85×10^{-14}
marine1	400,320	6,226,538	No	n/a	n/a	oom	oom	5.627s	6.25×10^{-14}
torso1	116,158	8,516,500	No	n/a	n/a	952.952ms	3.82×10^{-8}	2.018s	$\mathbf{3.75\times10^{-6}}$
atmosmodd	1,270,432	8,814,880	No	n/a	n/a	oom	oom	46.011s	2.48×10^{-16}
atmosmodl	1,489,752	10,319,760	No	n/a	n/a	oom	oom	40.869s	7.60×10^{-18}
memchip	2,707,524	14,810,202	No	n/a	n/a	26.259s	2.04×10^{-15}	8.046s	3.40×10^{-15}
Freescale1	3,428,755	18,920,347	No	n/a	n/a	30.025s	7.05×10^{-10}	10.174s	7.05×10^{-10}
rajat31	4,690,002	20,316,253	No	n/a	n/a	oom	oom	16.019s	9.31×10^{-15}
Transport	1,602,111	23,500,731	No	n/a	n/a	oom	oom	43.495s	3.81×10^{-10}
inline_1	503,712	36,816,342	Yes*	n/a	n/a	oom	oom	3.584s	2.77×10^{-15}
PFlow_742	742,793	37,138,461	Yes*	n/a	n/a	oom	oom	12.287s	4.17×10^{-11}
Emilia_923	923,136	41,005,206	Yes*	n/a	n/a	oom	oom	39.852s	5.27×10^{-23}
dielFilterV2	1,157,456	48,538,952	Yes	n/a	n/a	oom	oom	13.788s	1.45×10^{-11}
Flan_1565	1,564,794	117,406,044	Yes*	n/a	n/a	oom	oom	23.996s	9.85×10^{-16}
pres-cylin	1,711,464	133,562,188	Yes*	n/a	n/a	oom	oom	**2m6.195s**	4.90×10^{-13}
pmsim-hex8-27	157	6,581	Yes*	934.320µs	1.16×10^{-15}	736.383µs	2.85×10^{-16}	7.657ms	5.62×10^{-16}

Table 2: Calculations on Arch. Direct, UMFPACK and MUMPS. Complex-valued matrices.

				Direct		UMFPACK		MUMPS	
Matrix	**Nrow**	**NNZ**	**Sym**	**Total Time**	**Relative Error**	**Total Time**	**Relative Error**	**Total Time**	**Relative Error**
mhd1280b	1,280	22,778	No	717.206µs	1.31×10^{-15}	382.531µs	7.41×10^{-16}	5.759ms	1.17×10^{-15}
mplate	5,962	142,190	Yes	3.131s	4.66×10^{-11}	627.467ms	1.86×10^{-11}	109.002ms	6.28×10^{-11}
RFdevice	74,104	365,580	No	n/a	n/a	4.342s	1.22×10^{-13}	1.803s	1.52×10^{-7}
vfem	93,476	1,434,636	No	n/a	n/a	4.088s	4.70×10^{-8}	1.638s	6.54×10^{-8}
fem_filter	74,062	1,731,206	No	n/a	n/a	2.627s	6.61×10^{-11}	1.185s	1.40×10^{-10}
Chevron4	711,450	6,376,412	No	n/a	n/a	5.729s	1.60×10^{-11}	3.398s	4.16×10^{-11}
mono_500Hz	169,410	5,036,288	No	n/a	n/a	oom	oom	6.379s	5.64×10^{-9}
kim2	456,976	11,330,020	No	n/a	n/a	oom	oom	4.745s	2.73×10^{-18}
fem_hifreq_circuit	491,100	20,239,237	No	n/a	n/a	oom	oom	11.608s	1.00×10^{-10}
dielFilterV3clx	420,408	32,886,208	Yes	n/a	n/a	oom	oom	4.824s	1.68×10^{-11}

Table 3: Calculations on Arch. UMFPACK compared with the iterative solvers. Real-valued matrices.

				UMFPACK		BiCGStab		GMRES	
Matrix	**Nrow**	**NNZ**	**Sym**	**Total Time**	**Relative Error**	**Total Time**	**Relative Error**	**Total Time**	**Relative Error**
bwm2000	2,000	7,996	No	750.259µs	5.30×10^{-16}	28.781ms	8.72×10^{-15}	n/a	n/a
rdb5000	5,000	29,600	No	13.569ms	1.93×10^{-15}	n/a	n/a	25.734ms	3.77×10^{-12}
Goodwin_040	17,922	561,677	No	107.091ms	6.62×10^{-13}	1.477s	7.75×10^{-10}	n/a	n/a
fp	7,548	848,553	No	154.476ms	2.57×10^{-21}	n/a	n/a	n/a	n/a
xenon1	48,600	1,181,120	No	598.244ms	5.55×10^{-40}	2.732s	6.02×10^{-38}	2.935s	1.18×10^{-37}
twotone	120,750	1,224,224	No	337.133ms	1.13×10^{-14}	n/a	n/a	n/a	n/a
Raj1	263,743	1,302,464	No	oom	oom	n/a	n/a	n/a	n/a
boyd2	466,316	1,500,397	Yes	**1m18.248s**	2.86×10^{-13}	n/a	n/a	n/a	n/a
Goodwin_071	56,021	1,797,934	No	426.060ms	2.77×10^{-12}	11.724s	7.92×10^{-9}	n/a	n/a
darcy003	389,874	2,101,242	Yes	1.385s	6.95×10^{-12}	n/a	n/a	n/a	n/a
rma10	46,835	2,374,001	No	410.052ms	1.57×10^{-17}	n/a	n/a	n/a	n/a
helm2d03	392,257	2,741,935	Yes	1.683s	2.37×10^{-13}	n/a	n/a	n/a	n/a
stomach	213,360	3,021,648	No	1.781s	5.13×10^{-16}	n/a	n/a	n/a	n/a
oilpan	73,752	3,597,188	Yes*	494.931ms	1.26×10^{-15}	n/a	n/a	n/a	n/a
ASIC_680k	682,862	3,871,773	No	1.318s	2.20×10^{-11}	n/a	n/a	n/a	n/a
tmt_unsym	917,825	4,584,801	No	3.747s	5.96×10^{-8}	1.638s	1.91×10^{-7}	n/a	n/a
Goodwin_127	178,437	5,778,545	No	1.173s	6.03×10^{-12}	n/a	n/a	n/a	n/a
pre2	659,033	5,959,282	No	oom	oom	n/a	n/a	n/a	n/a
marine1	400,320	6,226,538	No	oom	oom	n/a	n/a	n/a	n/a
torso1	116,158	8,516,500	No	952.952ms	3.82×10^{-8}	n/a	n/a	n/a	n/a
atmosmodd	1,270,432	8,814,880	No	oom	oom	7.780s	7.66×10^{-15}	14.514s	6.88×10^{-15}
atmosmodl	1,489,752	10,319,760	No	oom	oom	5.751s	9.12×10^{-16}	7.781s	1.59×10^{-15}
memchip	2,707,524	14,810,202	No	26.259s	2.04×10^{-15}	21.341s	1.73×10^{-11}	8.059s	2.65×10^{-11}
Freescale1	3,428,755	18,920,347	No	30.025s	7.05×10^{-10}	7.383s	1.64×10^{-9}	n/a	n/a
rajat31	4,690,002	20,316,253	No	oom	oom	37.987s	4.75×10^{-12}	n/a	n/a
Transport	1,602,111	23,500,731	No	oom	oom	6.315s	1.90×10^{-9}	6.188s	6.29×10^{-10}
inline_1	503,712	36,816,342	Yes*	oom	oom	n/a	n/a	n/a	n/a
PFlow_742	742,793	37,138,461	Yes*	oom	oom	n/a	n/a	n/a	n/a
Emilia_923	923,136	41,005,206	Yes*	oom	oom	n/a	n/a	n/a	n/a
dielFilterV2	1,157,456	48,538,952	Yes	oom	oom	n/a	n/a	n/a	n/a
Flan_1565	1,564,794	117,406,044	Yes*	oom	oom	n/a	n/a	n/a	n/a
pres-cylin	1,711,464	133,562,188	Yes*	oom	oom	n/a	n/a	n/a	n/a
pmsim-hex8-27	157	6,581	Yes*	736.383µs	2.85×10^{-16}	818.326µs	8.57×10^{-14}	736.254µs	1.53×10^{-13}